

PUBLIC TECHNICAL PROFILE

Dr. Jonathan Meyer

Space environmental-test, qualification, and test-data systems. I build the facilities, instrumentation, procedures, data visibility, and evidence flow that turn demanding campaigns into a repeatable engineering and production function.

jonathan@jmeyer.spacejmeyer.space[Gossamer on GitHub](#)[PDF version](#)

This public-safe profile is for technical collaborators, space-hardware teams, and serious conversations around environmental test, qualification, production readiness, and modern test-data systems. It is not an application-specific CV.

Current Focus

- Environmental-test and qualification ownership for space hardware.
- TVAC, thermal cycling, vibration, shock, vacuum, cryogenic, and supporting infrastructure.
- Test capability as production infrastructure: readiness, repeatability, safety, throughput, documentation, and evidence.
- Early derisking, design-for-test feedback, requirement testability, and formal qualification.
- DUT telemetry, facility state, test-bed data, anomalies, source provenance, and requirement evidence.
- AI-assisted engineering workflows where they improve traceability and report drafting under human technical review.

Professional Experience

Mynaric Lasercom GmbH - Environmental Testing and Qualification

2019 to present, Munich area | Space Simulation Test Engineer -> Lead Engineer, Environmental Testing -> Team Lead, Environmental Testing and Qualification

- Built and scaled environmental-test capability for laser communication space hardware during startup-to-production growth.
- Worked across TVAC, thermal, vibration, shock, cleanroom-adjacent operations, instrumentation, safe execution, anomaly handling, and qualification evidence.
- Supported production ramp-up with flexible setups, fixtures, instrumentation, automation, and practical evidence discipline.
- Owned and supported major infrastructure work across thermal chambers, LN2 supply/exhaust, vacuum systems, oxygen-depletion safety concepts, dual-zone thermal control, and supplier interfaces.
- Built test-data workflows that connect DUT and test-bed sources with live visibility, requirement checks, and measurement-to-proof traceability.

TU Kaiserslautern - Research Associate / PhD Candidate

2012 to 2018

- Designed and built cryogenic and vacuum instrumentation for low-temperature gas-phase spectroscopy.
- Worked with 4 K cryogenic systems, vacuum vessels, ion-trap integration, electronics integration, drawings, fabrication coordination, and diagnostics.
- Developed ion-source, UHV, and high-voltage experience useful for testability discussions around vacuum, plasma, and electrically stressed subsystems.

Public Demonstrator: Gossamer

Gossamer is an independent, public-safe synthetic environmental-test demonstrator. It uses fictional spacecraft data to show an operating model for campaign state, source provenance, DUT and test-bed data visibility, command authority, anomalies, requirements traceability, live evidence, and evidence reports.

The project exists to discuss test-system architecture and operator workflows without exposing employer data, lab procedures, real telemetry, proprietary protocols, or private identifiers.

github.com/egidinas/gossamer

Selected Capabilities

TVAC

thermal cycling

vibration

shock

LN2 infrastructure

vacuum systems

ODH safety

test planning

qualification evidence

DUT telemetry

requirement traceability

Python analysis

LabVIEW

Siemens PLC

UHV/HV

ion-source characterization

supplier coordination

design-for-test

Education

Dr. rer. nat. in Physical Chemistry, TU Kaiserslautern

Dissertation in gas-phase spectroscopy and complex vacuum/cryogenic instrumentation.

Diploma in Chemistry, TU Kaiserslautern

Thesis in specialty ion-source development and characterization, including UHV/HV operation, diagnostics, and practical hardware iteration.

Languages: German native, English full professional, Mandarin conversational.

Public-safe profile. No employer data, lab procedures, real telemetry, proprietary protocols, or private identifiers are included.